σ-ADDITIVE PROBABILITY ON ORTHOMODULAR LATTICES: A SURVEY AND AN OPEN PROBLEM

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Abstract. For a Boolean algebra, the passage from a finitely additive, finitely coherent assignment of probabilities to a countably additive measure is governed by a single condition (countable additivity) and executed by classical machinery (Carathéodory; Loomis–Sikorski; Stone duality). On a *non-distributive* orthomodular lattice (OML) this machinery breaks, and the question of when finite coherence forces countable behaviour is open. This note surveys the problem and isolates a precise open question. After fixing the apparatus (orthomodular lattices, the gap between orthogonality and meet-zero, the state/meet-additive/charge ladder, the McDonald–Bimbó dual space of filters), we separate the question into two axes that the Boolean case fuses. The *extension* axis — does a state extend to a charge on the dual? — is a classical Horn–Tarski feasibility question and is settled negatively for $L(H)$ by an elementary finite construction. The *descent* axis — does such a charge concentrate on the physical points? — is open, and we locate it precisely. The combinatorial route to a witness (concrete, σ-orthocomplete, set-representable OMLs) is shown not to produce one: the interleaving property that would make σ-additivity exceed finite additivity on a concrete non-Boolean lattice is already satisfied by the product $\prod_n \mathrm{MO}_2$, which is concrete, σ-complete, non-Boolean and infinite, yet has its non-distributivity confined to finite blocks (*segregated*) and so fails to be a witness. The structural reason is that a faithful set representation (σ-tribe) of an OML has a distributive image and hence forces the lattice Boolean (Theorem 5.2), so no measure with no hidden realisation can arise from a faithful descent to points; the forcing boundary to Boolean is *subadditivity* of a unital set of states (Pták–Pulmannová [33]), not σ-additivity. Taking "no hidden realisation" to mean that the measure is not a mixture of dispersion-free states, this is equivalent to the state being *contextual* — not in the closed convex hull of the lattice's dispersion-free states (Lemma 5.3, in the lineage of Fine, Pitowsky and Abramsky–Brandenburger). The open problem is then sharp: **does a non-Boolean, σ-complete, concrete OML carry a σ-additive contextual state whose contextuality is σ-essential** — witnessed by no finite sub-OML? The finite version is already inhabited: Wright's pentagon ([40]) is a finite concrete OML carrying a non-classical $\{0, \frac{1}{2}\}$-valued state outside the dispersion-free hull, so the general spanning statement is false and only the σ-essential refinement remains. That refinement is a *compactness failure* of the same type as the non-axiomatizability of countable additivity over finitely additive coherence. The question is well-posed and has no construction in hand; recent work on coherent probabilities over Dynkin-systems (Derr–Williamson [9]) settles it *negatively under a Polish/Borel-regularity hypothesis*, leaving as the residue a descriptive-set-theory question — whether a witness can live where the σ-additive dispersion-free state space is non-Polish (Remark 5.1). A positive answer would be a non-Boolean analogue of localic measure theory; a negative one a σ-essential impossibility theorem. We review what is known and what is ruled out, and close with the two exits.

1. Introduction

All measurement is finite, yet the frameworks that organise empirical knowledge — probability theory, quantum mechanics — rest on infinite structures. The passage from finite observation to infinite structure is not automatic; it requires a commitment no finite body of evidence can compel. In probability theory the locus of that commitment is *countable additivity*:

$$A_1 \supseteq A_2 \supseteq \cdots, \quad \bigcap_n A_n = \varnothing \quad \Longrightarrow \quad \mu(A_n) \to 0.$$

De Finetti [8] held that only finite additivity is empirically grounded; Kolmogorov [26] adopted σ-additivity as an axiom.

The same tension recurs once observations need not be jointly performable. An observation *context* — a set of measurements performable together — is Boolean; gluing contexts along shared observations (a categorical colimit) returns:

$$\begin{aligned} \text{contexts mutually compatible} &\Rightarrow \text{Boolean algebra,} \\ \text{incompatible} &\Rightarrow \text{non-distributive OML.} \end{aligned}$$

(Gunji et al. [18]). The OML is *forced* by the context structure, not postulated. Incompatibility — quantum complementarity, but equally contextual experimental design or interfering sensors — drives the algebra out of the Boolean world. One caveat fixes the scope: for non-quantum examples the incompatibility must be *operationally imposed* (contexts genuinely non-corealizable, no joint probability space); it cannot arise intrinsically from one classical system's dynamics, since any finite family of classical observables on a common space jointly distributes (Kolmogorov), hence stays Boolean — contextuality there is failure of a global section (Abramsky–Brandenburger [1]; Fine [13]). The closed subspaces of a Hilbert space H form the prototype $L(H)$, not the premise.

On a non-distributive OML, Carathéodory's outer measure — needing a Boolean algebra of sets — does not apply, and the rescue is local to $L(H)$:

- For $L(H)$, Gleason's rigidity theorem [16] rescues it; Hilbert space is special and no general-OML analogue is known.
- Non-distributivity *per se* does not remove the 2-valued homomorphisms the Boolean engine needs — MO_3 is non-distributive yet carries them (§2.1, §4) — but it removes the *guarantee*; for $L(H)$, $\dim \geq 3$ they fail outright (Kochen–Specker [25]).

Whether enough survive is a question of concreteness, not non-distributivity — where an OML is *concrete* (Definition 2.7) when its elements can be faithfully modelled as honest *sets of outcomes*, equivalently when it carries an order-determining set of two-valued states (Gudder [17]); "enough survive" is then almost the definition. This is the qualifier on which the open problem turns (§4, §5).

The organising split. This note asks the shared question — *when does finite coherence force countable behaviour?* — on an OML. The vague question splits into two axes that the Boolean case fuses, and the survey is built around the split:

Axis	**Asks**	**Status**
Extension	Does a state extend to a charge on $S_0(A)$?	Settled (on $L(H)$, none does).
Descent	Does the charge concentrate on the physical points?	Open.

The relational standpoint. Two established moves motivate the approach:

- De Finetti [8] — probability without a presupposed sample space: coherent commitments about events, not measures on a given Ω.
- Birkhoff–von Neumann — propositions as a lattice (the OML of §2), not as subsets of a phase space.

Combining them, we start from the propositions and their entailment order and ask what that relational structure alone determines — the measure-theoretic counterpart of pointless (localic) topology. The dual is then not a space of points but the McDonald–Bimbó space of *filters* (§2.3), and a positive answer to the open problem would be a *relational* σ-additive probability theory on a non-distributive lattice — the non-Boolean analogue of localic measure theory. (The §5 statement makes "relational" precise as a *contextual* state, no hidden realisation; developed further in [27].)

2. Preliminaries

We collect the lattice-theoretic apparatus; all notions are standard.

2.1. Orthomodular lattices, and the orthogonal/meet-zero gap.

Definition 2.1 (Orthocomplemented lattice)**.** A *bounded lattice* is a partially ordered set in which every pair a, b has a join $a \vee b$ and meet $a \wedge b$, with greatest element $\mathbf{1}$ and least element $\mathbf{0}$. An *orthocomplementation* is a map $a \mapsto a^{\perp}$ satisfying, for all a, b:

(i) $a \wedge a^{\perp} = \mathbf{0}$ and $a \vee a^{\perp} = \mathbf{1}$ (complement),
(ii) $a^{\perp\perp} = a$ (involution),
(iii) $a \leq b \Rightarrow b^{\perp} \leq a^{\perp}$ (order-reversing).

Definition 2.2 (Incomparable; meet-zero; orthogonal)**.** For elements a, b of an orthocomplemented lattice, three relations of increasing strength:

- *incomparable*: $a \not\leq b$ and $b \not\leq a$ — neither entails the other (an order condition);
- *meet-zero*: $a \wedge b = \mathbf{0}$ — no common refinement;
- *orthogonal*, written $a \perp b$: $a \leq b^{\perp}$ (equivalently $b \leq a^{\perp}$) — one entails the other's negation, *decisively incompatible.*

Each implies the one above it ($a \perp b \Rightarrow a \wedge b = \mathbf{0} \Rightarrow$ incomparable, for $a, b \notin \{\mathbf{0}, \mathbf{1}\}$). For two *distinct atoms*, incomparable and meet-zero coincide

— an atom has only $\mathbf{0}$ below it, so $a \wedge b \in \{\mathbf{0}, a\}$, and $a \wedge b = a$ would force $a \leq b$; orthogonality remains strictly stronger.

Remark 2.3 (Meet-zero versus orthogonal). The implication $a \perp b \Rightarrow a \wedge b = \mathbf{0}$ reverses in every Boolean algebra, but fails once the lattice is non-distributive: there meet-zero is strictly weaker than orthogonality. The distinctions developed below all turn on this separation.

Definition 2.4 (Orthomodular lattice)**.** An orthocomplemented lattice A is *orthomodular* if

$$a \leq b \quad \Longrightarrow \quad b = a \vee (a^{\perp} \wedge b). \tag{1}$$

A *Boolean algebra* is the special case where $\wedge$ distributes over $\vee$; every Boolean algebra is orthomodular, not conversely. The motivating non-Boolean example is $L(H)$, the closed subspaces of a Hilbert space with $a^{\perp}$ the orthogonal complement.

Definition 2.5 (Completeness notions)**.** Two countable-completeness conditions on an OML A are distinguished throughout, the second strictly weaker. Here $\bigvee_n a_n$ denotes the *join* (least upper bound) of $\{a_n\}$ in the order of A, unique when it exists; "$\exists \bigvee_n a_n \in A$" asserts that it does.

- A is *σ-complete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A : \quad \exists \bigvee_n a_n \in A$$
(equivalently every countable family has a meet, by $\bigwedge_n a_n = (\bigvee_n a_n^{\perp})^{\perp}$);
- A is *σ-orthocomplete* if
$$\forall \{a_n\}_{n\in\mathbb{N}} \subseteq A \ \big(a_i \perp a_j \text{ for } i \neq j\big) : \quad \exists \bigvee_n a_n \in A.$$

The orthogonal families are a subclass of all countable families, so σ-completeness quantifies over a strictly larger domain and is the stronger hypothesis: σ-completeness $\Rightarrow$ σ-orthocompleteness. Only the weaker condition is needed below. For a state s, σ-additivity is the requirement that

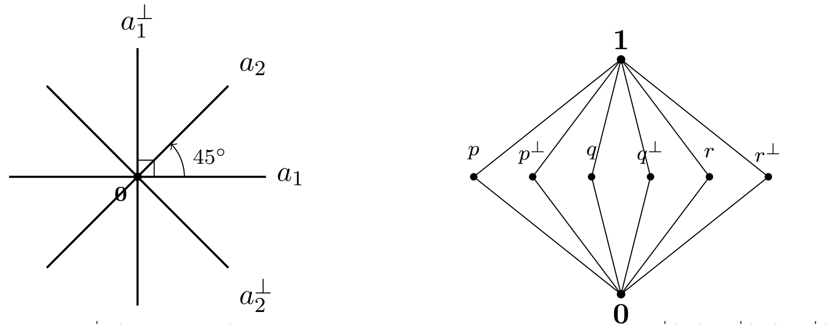
$$s\Big(\bigvee_n a_n\Big) = \sum_n s(a_n)$$

for every countable pairwise-orthogonal family $\{a_n\}$. Its left side invokes $\bigvee_n a_n$ *only* for orthogonal families — so σ-orthocompleteness is exactly the hypothesis that makes it well-posed. Full σ-completeness is the stronger condition manipulated by the Loomis–Sikorski representation (§2.4).

Example 2.6 (MO_n, and the gap made concrete)**.** MO_n is the lattice with $\mathbf{0}$, $\mathbf{1}$, and n complementary pairs of incomparable atoms, all sharing only $\mathbf{0}$ and $\mathbf{1}$. The geometric model is n lines through the origin: each line a is orthogonal to its perpendicular $a^{\perp}$, while two atoms from distinct pairs are non-perpendicular lines. It is orthomodular and non-distributive ($n \geq 2$): two atoms drawn from *distinct* complementary pairs have

$$a \wedge b = \mathbf{0} \quad \text{(meet only at the origin)} \qquad \text{yet} \qquad a \not\leq b^{\perp},$$

so meet-zero $\neq$ orthogonal already here. MO_3 is the smallest non-distributive OML.



$a_1 \perp a_1^\perp$ (right angle), $a_1 \not\perp a_2$ but $a_1 \wedge a_2 = \mathbf{0}$

complementary pairs $(p, p^\perp), (q, q^\perp), (r, r^\perp)$; any two non-paired atoms meet at $\mathbf{0}$, join at $\mathbf{1}$

FIGURE 1. MO_n illustrated. *Left:* the geometric model for MO_2 — four atoms as lines through the origin in $\mathbb{R}^2$. The pair $a_1, a_1^\perp$ meets at a right angle (orthogonal), whereas a_1, a_2 meet at $45°$: their lattice meet is still $\mathbf{0}$ (only the origin in common), yet they are *not* orthogonal — the meet-zero $\neq$ orthogonal gap. *Right:* the Hasse diagram of MO_3, the smallest non-distributive OML: three complementary pairs of incomparable atoms between $\mathbf{0}$ and $\mathbf{1}$.

Definition 2.7 (OMP; order-determining states; concrete logic)**.** An *orthomodular poset* (OMP) weakens Definition 2.4: joins $a \vee b$ are required only for *orthogonal* pairs. A set S of states on A is *order-determining* if it separates the order, i.e.

$$\big(s(a) \le s(b) \text{ for all } s \in S\big) \implies a \le b.$$

An OMP is a *concrete logic* (equivalently *set-representable*) if it embeds into $(\mathcal{P}(X), \subseteq, X \setminus (-))$ for some set X, preserving existing orthogonal joins — i.e. its propositions are modelled as actual *sets of outcomes*. Concreteness is equivalent to admitting an order-determining set of *two-valued* states (Gudder [17]), and it is the property that separates the open frontier from the settled cases (§5).

Remark 2.8 (Concreteness does not close the gap). A set-representable OMP needs only closure under complement and *disjoint* union [4]; intersection-closure would force a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L-member inside $A \cap B$, so

$$\underbrace{A \cap B = \varnothing}_{\text{orthogonal}} \implies \underbrace{a \wedge b = \mathbf{0}}_{\text{meet-zero}}, \qquad \text{but not conversely.}$$

Concreteness does not force the converse: MO_3 is itself concrete (Definition 2.7; Gudder [17]), so the paradigmatic meet-zero-$\neq$-orthogonal example is already a concrete logic. What closes the gap is a richness/atomicity condition, not concreteness:

$$a \wedge b = \mathbf{0} \;\Rightarrow\; a \perp b,$$

equivalently: every nonempty $A \cap B$ contains a nonzero L-member — the abstract form being Tkadlec's *Boolean orthoposet* condition [36] (which does *not* mean Boolean *algebra*).

2.2. **The state/meet-additive/charge ladder.**

Definition 2.9 (State; meet-additive state; charge-extendible)**.** Let A be an OML. A *state* is a map $s : A \to [0, 1]$ with $s(\mathbf{1}) = 1$ that is additive on *orthogonal* pairs: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$; that is, a probability assignment whose additivity is guaranteed only on decisively incompatible propositions. Three strengthenings, differing in which pairs additivity is demanded on and strictly increasing in general (Example 2.10 on MO_3):

(1) *State*: $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$.
(2) *Meet-additive*: $s(a \vee b) = s(a) + s(b)$ whenever $a \wedge b = \mathbf{0}$ (binary, strictly stronger — it reaches the gap pairs of Definition 2.2). We avoid the word "valuation" here: in lattice theory it denotes the modular function $v(a)+v(b) = v(a\wedge b)+v(a\vee b)$, a different condition.
(3) *Charge-extendible*: $\sum_i s(a_i) \le 1$ for every pairwise-meet-zero family $\{a_i\}$ ($a_i \wedge a_j = \mathbf{0}$, $i \ne j$; strictly stronger again). This is the condition necessary for extension to a charge on the dual space (Definition 2.13) and the *extension* condition of §3.

Example 2.10 (The ladder is strict)**.** On MO_3 with complementary pairs $(p, p^\perp)$, $(q, q^\perp)$, $(r, r^\perp)$, the maximally mixed $s \equiv \frac{1}{2}$ is meet-additive (clears (2)) yet fails (3) at $\{p, q, r\}$: $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So no state on MO_3 is charge-extendible.

Definition 2.11 (Normal and singular states on $L(H)$)**.** On $A = L(H)$ a state s is

- *normal* if it is completely additive on every orthogonal family of projections,

$$s\Big(\bigvee_{i\in I} p_i\Big) = \sum_{i\in I} s(p_i) \quad \text{for all (not only countable) orthogonal } \{p_i\},$$

  equivalently $s(\cdot) = \mathrm{tr}(\rho\,\cdot)$ for a density operator ρ (Gleason [16], dim$\geq$3);
- *singular* (purely) if

$$s(e) = 0 \quad \text{for every finite-rank projection } e.$$

Every state decomposes into a normal and a singular part (Takesaki); the two classes are exactly those distinguished by the arguments of §3.

2.3. **The McDonald–Bimbó dual space.**

Definition 2.12 (Dual space; representation map)**.** The McDonald–Bimbó duality [28] assigns to an OML A a compact space

$$S_0(A) = \big(F(A), \subseteq, \perp_A, P(A), T\big),$$

where $F(A)$ is the set of *filters* of A, ordered by $\subseteq$ and carrying an orthogonality relation $\perp_A$ and a Stone topology T, and $P(A) = \{\,{\uparrow}p : p \in A\,\} \subseteq F(A)$, with ${\uparrow}p = \{a \in A : a \geq p\}$, is the set of *principal* filters — the "physical points," the realisation datum (canonical, unlike the non-canonical pure points of the Boolean case). For $a \in A$ set $h(a) = \{x \in F(A) : a \in x\}$; then h is an OML isomorphism onto the $\perp$-stable clopens $\mathrm{CO}(S_0(A))^\dagger$. Define $\mu(h(a)) = s(a)$. Here $h(a)$ is the set of points at which a holds and μ is the state transported onto those sets; the open problem is whether μ is preserved under countable operations on them.

Definition 2.13 (Charge)**.** A *charge* on $S_0(A)$ is a map $\nu : \mathrm{CO}(S_0(A)) \to [0,1]$ on the *full* Boolean algebra of clopen subsets (not merely the $\perp$-stable ones in the image of h), with

$$\nu(S_0(A)) = 1, \qquad \nu(U \cup V) = \nu(U) + \nu(V) \ \text{ whenever } U \cap V = \varnothing.$$

A state s *extends to a charge* if some such ν satisfies $\nu(h(a)) = s(a)$ for all $a \in A$. Because h preserves meets, a pairwise-meet-zero family in A maps to disjoint clopens, so a charge necessarily satisfies $\sum_i s(a_i) \leq 1$ over such families — the charge-extendibility condition of Definition 2.9(3).

Remark 2.14 (The duality is finitary)*.* The representation map h satisfies

$$h\Big(\bigvee_{i\leq n} a_i\Big) = \Big(\bigcup_{i\leq n} h(a_i)\Big)^{\perp\perp} \quad \text{(finite)},$$

$$h\Big(\bigvee_{n} a_n\Big) \neq \Big(\bigcup_{n} h(a_n)\Big)^{\perp\perp} \quad \text{(countable)}:$$

finitary OML homomorphisms need not preserve infinite suprema. This is the central obstruction the open problem of §5 confronts, and it is self-dual: since $a \mapsto a^\perp$ is an order anti-isomorphism, $\bigwedge_n a_n = (\bigvee_n a_n^\perp)^\perp$, so the failure for countable meets is the De Morgan dual of the failure for countable joins. The rival duality of Cannon–Döring [5] is also finitary and carries no measure; we use McDonald–Bimbó because it carries $P(A)$ as explicit structure.

2.4. **The Boolean engine: Stone duality.**

In the Boolean case the descent equivalence

$$\sigma\text{-additive} \quad \Longleftrightarrow \quad \text{the measure concentrates on the physical points}$$

is proved through *Stone duality*, in four steps [27]:

(S1) a Boolean event algebra B embeds in the clopen algebra $\mathrm{CO}(\mathrm{St}\,B)$ of its Stone space;

(S2) a charge on B extends to a Baire measure on $\mathrm{St}\,B$ (Carathéodory);

(S3) σ-additivity $\iff$ the measure vanishes on meagre Baire sets (Rao–Rao [34]);
(S4) meagre-null $\iff$ the measure lives on the ultrafilters closed under countable intersection = the realised points.

Steps (S3)–(S4) are the measure-theoretic face of the *Loomis–Sikorski* representation,

$$\sigma\text{-complete Boolean algebra} \;\cong\; (\sigma\text{-tribe of sets}) \,/\, (\sigma\text{-ideal}),$$

the countable-join bookkeeping that drives the equivalence. The contrast with the OML case is exact: the Boolean engine (Stone duality together with the Loomis–Sikorski representation) has *no* known OML analogue (§4.3). Supplying such an engine, or proving none exists, is the open problem of §5.

Remark 2.15 (Why the bridge breaks: meet-closed but not join-prime). A "realised" point x carries two conditions, which Rao–Rao's "vanishes on meagre sets" makes coincide in the Boolean case:

(M) closed under countable meets;
(J) $x \ni \bigvee_n a_n \Rightarrow x \ni a_n$ for some n (off the join-defect).

Non-distributivity separates them. For a principal filter $x = \uparrow p$, condition (M) always holds, whereas (J) can fail: $p \le \bigvee_n a_n$ does not force $p \le a_n$ for some n. In $L(H)$ this is witnessed by $a_n = \mathrm{span}(e_n)$ and $p = \mathrm{span}(e_1 + e_2)$:

$$p \;\le\; \bigvee_n a_n = H, \qquad \text{yet} \qquad p \;\not\le\; a_n \;\text{ for every } n.$$

So $\uparrow p$ is a realised point lying inside the join-defect $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$. Topologically $\uparrow p$ is *isolated* in $S_0(L(H))$, hence:

$$D \text{ non-meagre (OML)} \qquad \text{vs.} \qquad D \text{ nowhere dense (Boolean)}.$$

This is the finitary obstruction of Remark 2.14 on the dual: the category route (clopens modulo the meagre ideal) fails on the same D as the measure route, and reduces to the MacNeille completion of §4.3(ii).

3. Two axes: extension and descent

The central question — *when does a state s on an OML A extend to a σ-additive measure on $S_0(A)$?* — splits into two **independent** axes that the Boolean case fuses.

Extension.:
Does s extend to a finitely additive charge on the *full* Boolean clopen algebra of $S_0(A)$? This is governed by the meet-zero/orthogonal gap (§2.1) and is *settled*: it is a Horn–Tarski/Pitowsky feasibility question, a decidable linear program in the finite case, and can fail for every state.

Descent.:
Once extended, does the charge concentrate on the physical points $P(A)$? This is governed by σ-additivity — it needs the missing engine (§2.4) — and is *open*.

The extension axis is the abstract form of a familiar question: extending a finitely-additive assignment on incompatible propositions to a single joint distribution is exactly the feasibility problem behind the Bell inequalities (Fine [13]: a Bell model exists iff the correlations admit a joint distribution; Pitowsky [31]: the Bell inequalities are the facets of a correlation polytope), and Bell and Kochen–Specker non-classicality are jointly an extension problem for partial Boolean algebras (Budroni–Morchio [2], with the necessary-and-sufficient conditions of Horn–Tarski [22]). What is distinctive here is only that the extension is sought against the dual $S_0(A)$ rather than within a fixed scenario; the obstruction is the same finitary one, and it is settled. Two observations pin it down.

Proposition 3.1 (Finite case)**.** *For finite A the extension condition is a decidable linear program (Horn–Tarski* [22] */ Pitowsky correlation-polytope feasibility* [31]*; Budroni–Morchio* [2] *for the partial-Boolean-algebra formulation); since h preserves meets, meet-zero elements map to disjoint clopens and a charge must satisfy $\sum_i s(a_i) \leq 1$ over pairwise-meet-zero families. It can fail for* every *state (*MO_3*, Example 2.10).*

Proposition 3.2 (Infinite $L(H)$, all states)**.** *Let $\dim H = \infty$.* No *state on $L(H)$ — normal or singular — extends to a charge on $S_0(L(H))$.*

Proof. The argument embeds a copy of MO_2 in $L(H)$ whose four atoms are infinite-dimensional, so that the obstruction reaches the singular states as well. Write $H = H_0 \otimes \mathbb{C}^2$ with H_0 infinite-dimensional, and fix an orthonormal basis e, f of $\mathbb{C}^2$. Set

$$a_1 = H_0 \otimes \mathbb{C}e, \quad a_1^{\perp} = H_0 \otimes \mathbb{C}f, \quad a_2 = H_0 \otimes \mathbb{C}(e+f), \quad a_2^{\perp} = H_0 \otimes \mathbb{C}(e-f).$$

These four infinite-dimensional subspaces form a copy of MO_2 in $L(H)$:

$$\begin{aligned} &a_1 \perp a_1^{\perp}, \quad a_2 \perp a_2^{\perp}, \quad a_i \vee a_i^{\perp} = H; \\ &a_i \wedge a_j = \mathbf{0} \ \text{ but } \ a_i \not\perp a_j \ \text{ (cross pairs)}. \end{aligned}$$

Orthoadditivity with $s(\mathbf{1}) = 1$ gives $s(a_i) + s(a_i^{\perp}) = 1$ for $i = 1, 2$, using only additivity on orthogonal pairs, so it holds for *every* state. The family $\{a_1, a_1^{\perp}, a_2, a_2^{\perp}\}$ is pairwise meet-zero, and since h preserves meets ($h(a \wedge b) = h(a) \cap h(b)$), its members map to disjoint clopens; hence any charge satisfies $\sum_i s(a_i) \leq 1$. The two counts collide:

$$\sum_i s(a_i) = 2 \ \text{ (orthoadditivity)} \qquad \text{versus} \qquad \sum_i s(a_i) \leq 1 \ \text{ (any charge)}.$$

The construction is finite ($n = 4$); σ-completeness of the lattice plays no role. □

The MO_2 count is standard (orthoadditivity together with $s(\mathbf{1}) = 1$; see [23]) and $\sum s(a_i) = 2$ is already state-independent in $L(\mathbb{C}^2)$. The role of the *infinite-dimensional* realisation is to extend the obstruction to the singular states ($s(e) = 0$ on every finite-rank e):

Atoms of MO_2	$s(a_i) + s(a_i^\perp)$	**Reaches**
lines in $L(\mathbb{C}^2)$ (finite rank)	$= 1$ only if s charges finite rank	normal states only
$H_0 \otimes (\cdot)$, $\dim H_0 = \infty$	$= 1$ regardless of finite-rank behaviour	all states, incl. singular

Keeping each atom infinite-dimensional drops the finite-rank precondition of the line argument (Remark 3.3), and that is what closes the extension axis on $L(H)$ completely.

Remark 3.3 (An alternative argument for normal states only). A second, technique-disjoint argument reaches the normal states without the MO_2 family. If $s(e) > 0$ on some finite-dimensional e, it is positive on a 2-plane e_0; taking k pairwise-non-orthogonal lines $p_1, \ldots, p_k \subset e_0$ yields $2k$ pairwise-meet-zero lines with $s(p_i) + s(p_i^\perp) = s(e_0)$, so a charge forces

$$k\, s(e_0) \leq 1 \text{ for all } k \quad \Longrightarrow \quad s(e_0) = 0,$$

contradicting $s(e_0) > 0$. Lifting s to a functional via Mackey–Gleason / Bunce–Wright [3] ($B(H)$ has no type I_2 summand) and applying the Takesaki normal/singular decomposition identifies which states the argument reaches: it covers states with a nonzero normal part ($s(e) > 0$ for some finite-rank e), including every normal (Gleason) state, and misses the purely singular states ($s(e) = 0$ for all finite-rank e; e.g. ultrafilter vector states or Calkin-algebra pullbacks). Proposition 3.2 subsumes the missed case; this argument is recorded because its technique is independent and locates the singular states precisely.

4. What is known

The extension axis is settled — on $L(H)$ no state extends (Proposition 3.2); the descent axis turns on whether the Boolean engine of §2.4 has an OML analogue. Three things bear on that:

1. the Boolean benchmark such an analogue must match (§4.1);
2. the partial OML results, positive and obstructive (§4.2);
3. the routes already known to be blocked (§4.3).

4.1. **The Boolean benchmark.** The conditions an OML engine would have to reproduce. The Kelley–Vladimirov–Pták characterisation (Fremlin [14], Thm 391D) is complete: a Boolean algebra carries a strictly positive σ-additive measure iff it satisfies all three conditions below — each of which loses its meaning off the distributive world.

Boolean (KVP) condition	**OML analogue**
Dedekind σ-completeness	exists (σ-orthocompleteness)
weak (σ, ∞)-distributivity	none — relies on $\bigwedge$ distributing over $\bigvee$
chargeability	none — no structural characterisation on OMLs

4.2. **Positive and obstructive OML results. Positive** (require structure rich enough to approximate Hilbert-space geometry):

- Gleason [16] — $L(H)$, $\dim \geq 3$; σ-additivity *on the lattice*.
- Bunce–Wright [3] — JBW projection lattices; Chetcuti–Dvurečenskij [6].

Obstructive (Boolean machinery cannot be transplanted):

- Pták–Pulmannová [33] — a *finitary* condition (*subadditive*-unitality $s(a \vee b) \leq s(a) + s(b)$, no countable joins) already collapses the *lattice* to Boolean (their Thm 1); the OM*poset* version fails (Müller's set-representable counterexample [29]).
- Navara [30] — regularity does not force σ-additivity. Pták [32] — exotic state spaces via Greechie pasting.

4.3. **Three routes to a σ-OML engine, all unavailable.** The descent argument needs an OML analogue of the Boolean engine — a σ-Stone duality, or equivalently a Loomis–Sikorski representation. The three known routes are all unavailable, but not on the same footing: routes (i) and (ii) are closed by *theorem* (the hypothesis that would yield the representation provably forces Boolean, resp. leaves the category), whereas route (iii) is open in the strong sense — no construction is known, and no impossibility theorem rules one out (Remark 4.1). It is route (iii), restated, that is the Open Problem of §5.

(i) **RDP / effect-algebra.** Loomis–Sikorski holds for σ-MV algebras [11] and monotone σ-effect algebras *with* the Riesz Decomposition Property [12]; but a lattice effect algebra has RDP iff it is MV, and OML $\cap$ MV = Boolean (Kalmbach [23]), so the hypothesis that would yield the representation collapses the class:

$$\mathrm{OML} + \mathrm{RDP} \iff \text{Boolean}.$$

The *sharp-skeleton* evasion — representing the OML as the *sharp* elements of an ambient RDP effect algebra, so that only the ambient carries RDP — also closes: in any RDP effect algebra the sharp elements coincide with the center and are hence Boolean (Jenča 2001, Cor. 4.3; via Greechie–Foulis–Pulmannová 1995), so a non-Boolean sharp skeleton witnesses RDP-failure. Route (i) is closed against this refinement too (σ-completeness is not on the load path). The MacNeille completion of an OML need not be orthomodular (Harding [19]); one cannot complete to absorb countable joins. The topological version of this route — clopens of $S_0(A)$ modulo the meagre ideal — is this same completion in another form (the regular-open algebra), and collapses non-distributivity (Remark 2.15).

(ii) **σ-Stone duality.** None is *known*, and none is *ruled out*: the existing dualities are finitary (McDonald–Bimbó, Remark 2.14) or equational with no set/tribe representation (Freytes [15], on σ-complete OMLs), but no theorem forbids a countable-join-preserving one. The one machinery that supplies a point-free σ-additive valuation — topos/Bohrification — *sidesteps* this route rather than settling it (its σ-additivity lives on a distributive object / per-context Boolean blocks; see §4.5), so route (iii) stays open but the known candidate-machinery is exhausted as a crack.

4.4. **Prior art on point-free quantum probability.** Every existing point-free or directed-context construction fails at least one of the two requirements — σ-additivity and a natively non-distributive structure — as tabulated below.

Gleason tradition:
Varadarajan [37], Kalmbach [23], Pták–Pulmannová [33]: σ-additive on a non-distributive OML, but real-valued on a fixed lattice — not point-free.

Döring 2009: [10] point-free on the spectral presheaf over a directed poset — but provably only *finitely additive*, on a distributive Heyting algebra.

Bohrification (HLS):
[20, 21] point-free and the internal valuation *is* σ-additive (Scott-continuous) — but factors through an internally *distributive* locale, σ-additive only per commutative context. Non-distributivity is tamed, not carried natively.

Localic valuations:
Vickers [38], Simpson [35], Coquand–Spitters [7]: point-free σ-additive, but on *frames* — distributive by definition.

OML Stone dualities:
McDonald–Bimbó [28] and Cannon–Döring [5]: purely structural, no measure, no σ-version.

Each entry satisfies some of the requirements but not all. The open case is their conjunction, satisfied by no existing construction:

$$\boxed{\text{point-free} \times \sigma\text{-additive} \times \text{natively non-distributive} \times \text{directed-system-built.}}$$

(The directed-context-yields-non-distributivity construction is not new — it is the shape of Bohrification and of colimit-over-Boolean-contexts generation. Novelty would lie only in the σ-additive $\times$ point-free $\times$ native combination.)

Remark 4.1 (The boundary of the negative results). Two qualifications delimit what is ruled out.

(i): **No impossibility theorem covers the concrete class.** The known no-go results are finite or two-valued, and the one positive occupant is non-concrete:

$$L(H)\ (\dim \geq 3): \quad \text{no two-valued states at all (Kochen–Specker [25]),}$$

a fortiori none order-determining. Concreteness is the qualifier on which the open frontier turns.

(ii): **No σ-Loomis–Sikorski exists**, and this is visible in the dualities: both take infinite joins as a closure, not a union,

$$\text{Cannon–Döring: } \mathrm{cls}(\textstyle\bigcup S_i); \qquad \text{McDonald–Bimbó: } (\textstyle\bigcup h(a_n))^{\perp\perp}.$$

This is not to be overstated: Gudder's concrete logics [17] *are* set-representable non-Boolean orthoposets — but as point-ful posets, not via a σ-Loomis–Sikorski theorem.

4.5. **What the topos route does and does not buy.** The topos / Bohrification programme is the one machinery that supplies a genuinely point-free *and* σ-additive (directed-continuous) valuation on a quantum logic, so it is the natural candidate to crack route (iii). A mechanism-level read of the two principal constructions shows it is *the same wall, with tools*: the σ-additive valuation always lives on a *distributive* object, and the bridge to the non-distributive lattice routes its countable additivity through per-context Boolean blocks. The non-distributivity is relocated into base-poset variation and a coarse-graining map, never carried by the measure. This is the topos analogue of the universal floor (§5.1): a measure that meets the non-distributive structure faithfully forces it Boolean.
Bohrification (Heunen–Landsman–Spitters). A state on A becomes a *continuous probability valuation* $\mu : O(\Sigma(A)) \to [0,1]_l$ on the internal Gelfand spectrum $\Sigma(A)$, which is an internal *locale* — by construction a compact regular *frame* [21, §2, §6], hence internally distributive. The defining continuity axiom $\mu(\bigvee_i U_i) = \bigvee_i \mu(U_i)$ for directed families (Def. 6.11) is precisely the localic, Scott-continuous form of σ-additivity — but it is asked of $O(\Sigma(A))$, a frame, where $\bigvee$ is distributive. The non-commutativity of A sits entirely in the base poset $\mathcal{C}(A)$ of commutative subalgebras and in the topos's intuitionistic internal logic; the valuation never integrates over a non-distributive lattice.

Remark 4.2 (The HLS valuation does not reach the prize — objection and answer). Theorem 6.19 of [21] is a bijection between continuous probability valuations on $\Sigma(A)$ *and* probability measures on $\mathrm{Proj}(A)$ — the genuinely non-distributive object — so it *looks* like a σ-additive measure on a non-distributive OML. It is not a counterexample to the wall, for two independent reasons.

Primary (σ-essential).:
By Definition 6.17(a) a probability measure on the countably complete OML $\mathrm{Proj}(A)$ is a map that *restricts to a σ-Boolean-algebra morphism on every countably complete Boolean sublattice*. Its countable additivity is thus *defined block-by-block on the Boolean sub-structure* and glued by naturality over $\mathcal{C}(A)$; it never crosses a non-distributive (non-commuting) join. This is the *antithesis* of σ-essential contextuality (§5.1): the prize demands contextuality witnessed by the countable whole and by *no* Boolean sub-structure, whereas the topos route confines *all* of its countable additivity *inside* Boolean contexts. This argument is independent of concreteness.

Secondary (non-concrete).:
The non-distributive object here is $\mathrm{Proj}(A)$, the projection lattice of a (Rickart) C^*-algebra — the Gleason / $L(H)$-type lattice, non-concrete by Kochen–Specker (Rmk 4.1). So Bohrification *recasts the Gleason occupant* (the top row of §4.4) in topos language; it does not reach the concrete class on which the open frontier turns.

Döring–Isham (spectral presheaf). Measures live on the clopen subobjects $\mathrm{Sub}_{cl}(\Sigma)$, a complete Heyting algebra / locale — distributive, and *explicitly not* a σ-algebra [10]. The non-distributive $\mathrm{Proj}(H)$ enters only through *daseinisation* $\delta : \mathrm{Proj}(H) \to \mathrm{Sub}_{cl}(\Sigma)$, which is a coarse-graining, not a lattice homomorphism: outer δ^o preserves all joins but not meets, inner δ^i preserves all meets but not joins, and — in Wolters' words [39] — "it cannot preserve both, as $[\mathrm{Sub}_{cl}(\Sigma)]$ is distributive, whereas $[\mathrm{Proj}(H)]$ is nondistributive." This is the exact topos twin of the finitary-duality wall (Rmk 2.14). The measure is in general only *finitely* additive; the ceiling is driven by the state class (to cover type-III / KMS / non-normal states), and σ-additivity is recovered only *locally* — countable additivity for families pairwise orthogonal at a single context V, equivalently within one Boolean block, and only for normal states. As with HLS, the apparent "σ-additive on projections" statement holds only over pairwise-*orthogonal* (hence commuting, hence single-context, hence Boolean) families: the additivity never crosses a non-distributive join.

The f.a.-vs-σ diagnosis. The route lands on the programme's finitely-additive-versus-σ gap exactly: σ-additivity is available *precisely where the object is distributive* — the internal frame, or the per-context Boolean blocks where Loomis–Sikorski already applies ("the first arrow is free") — and degrades to finite additivity (Döring) precisely when a single global measure is demanded across the non-distributive whole. Of the four requirements (point-free $\times$ σ-additive $\times$ *natively non-distributive* $\times$ directed-system-built), the topos route satisfies three and misses exactly the one that is the crux: native non-distributivity. The machinery engages the problem; it does not hold a handle on it. It *sidesteps* the wall (an internally distributive Gelfand spectrum is a design goal, so constructive Gelfand duality goes through) rather than *settling* the σ-essential question — which stays open, neither inhabited nor closed.

5. The open problem

Open Problem. Is there a non-Boolean, σ-complete, *concrete* OML carrying a σ-additive *contextual* state — a state with no global hidden joint distribution, equivalently one not lying in the closed convex hull of the lattice's dispersion-free (two-valued) states — whose contextuality is *σ-essential*, i.e. witnessed by no finite sub-OML? Or can one prove that no such state exists?

Remark 5.1 (Relation to Derr–Williamson 2023 — status under assessment). The problem above is **not known to be plainly open**: it must be read against Derr–Williamson [9], which studies coherent probabilities on *pre-Dynkin-systems* (concrete orthomodular posets) and *Dynkin-systems* (concrete σ-complete OMLs) from the imprecise-probability side. Their Theorem 4.5 characterises finite-additive global extendability by a finite condition that matches "non-contextual on every finite sub-OML," and their Theorem D.6 upgrades this to a *countably additive* global extension *under the hypotheses* that the underlying set is Polish, the generated σ-algebra is its Borel algebra, and the σ-blocks are inner-regular. Where those hypotheses hold, the σ-essential cell is *empty* — no witness exists. Two identifications, however, remain to be verified against their definitions before the present problem can be declared (conditionally) settled, and each is a known trap of this subject:

(i) whether their "σ-extendable" (extends to a countably additive measure on the underlying point set) coincides with "non-contextual" here (lies in the closed convex hull of the *dispersion-free* states), which on a concrete logic form a strictly larger set than the point evaluations; and
(ii) whether their Polish hypothesis on the underlying *point set* is the same condition as Polishness of the *state space* S^{σ}_{df} — a σ-version of the (A)/(B) point-space equivocation of Remark 5.6.

Pending these, the honest status is: the finite-additive boundary (Theorem 4.5) is settled and agrees with the account below; the σ-essential cell is settled *negatively under Polish/Borel regularity*, with the residual question — whether a witness can live where S^{σ}_{df} is analytic-but-not-Borel / non-Polish — a descriptive-set-theory problem that subsumes the off-centre/non-band constraints of §6.

This is the relational-probability prize made precise. "Relational" means *no hidden realisation*: the measure must not be recoverable from positing an auxiliary space of definite states behind the propositions. Three point-spaces must be kept apart (the recurring trap of this subject): the *canonical* dual $P(A) = \{\uparrow p\}$ (one principal filter per *element*, §2.3); the *dispersion-free states* (two-valued states, one per consistent global *valuation* — the hidden-variable points); and the *external (A)-points* (Gleason's Hilbert rays, Remark 5.6). The canonical $P(A)$ is *permitted*; the objection is only to building the measure on a *smuggled* space — the (A)-points, or equivalently a global hidden-variable joint over the dispersion-free states. A state has a hidden realisation exactly when it is a mixture of dispersion-free states (Lemma 5.3), so a relational measure is precisely a *contextual* state, and the reduction below turns the existence question into this one. (Note this is a condition on the dispersion-free states, *not* on $P(A)$: a witness is point-rich on $P(A)$ yet has no global dispersion-free joint — the two spaces are different, which is exactly why "permitted" and "escaped" do not conflict.) (We do not pursue the stricter "strictly point-free / no canonical points either" reading: it makes concreteness itself the enemy and collapses against the universal that follows, so it offers no inhabitable target.)

5.1. **The reduction to contextuality, and what frames it.**

The universal floor. The naive route — realise the lattice as honest sets so that a measure becomes a measure on a space — is closed for every non-Boolean OML.

Theorem 5.2 (Universal floor)**.** *A* faithful set representation *of an OML L — an injective $h : L \to \mathcal{P}(\Omega)$ with $h(a \wedge b) = h(a) \cap h(b)$, $h(a \vee b) = h(a) \cup h(b)$, $h(a^{\perp}) = \Omega \setminus h(a)$ (a σ-tribe / Loomis–Sikorski representation when σ-joins are preserved) — exists only if L is Boolean.*

Proof sketch. Set operations $\cap, \cup, {}^{c}$ on $\mathcal{P}(\Omega)$ distribute, so the image $h(L)$ is a distributive sublattice; h injective and lattice-preserving makes L isomorphic to a distributive ortholattice, i.e. Boolean. ☐

Hence no relational measure can arise from a faithful descent to points. What a concrete OML *does* have (Definition 2.7) is an order-determining family of dispersion-free states; the live question is whether every state is a mixture of them.

The reduction. The following equivalence reduces "no hidden realisation" to a spanning condition on the dispersion-free states. Its content is not new — for a Bell scenario it is Fine's theorem [13] (a joint distribution exists iff a deterministic hidden-variable model does), and the convex-geometric form is Pitowsky's correlation-polytope picture [31]; the formulation for a general (concrete) OML, with dispersion-free states as the deterministic vertices, is the natural lift, in the lineage of the sheaf-theoretic account of contextuality [1].

Lemma 5.3 (Relational = contextual)**.** *Let L be a concrete OML with dispersion-free state space S_{df}, and w a state on L. Then w admits a hidden realisation — a probability measure on a space of global two-valued valuations of L inducing w — if and only if $w \in \overline{\mathrm{conv}}\,(S_{\mathrm{df}})$. We call w* contextual *when it is not.*

Proof sketch. A global two-valued valuation of L is exactly a dispersion-free state, so a hidden realisation is a probability measure μ on S_{df} with $w(a) = \int \omega(a)\, d\mu(\omega)$ for all a — i.e. w is the barycentre of μ. The barycentres of probability measures on the (weak-$*$ compact) set S_{df} are exactly $\overline{\mathrm{conv}}(S_{\mathrm{df}})$. The handle is *spanning*, not simplicity: non-uniqueness of μ is irrelevant; only its *existence* matters. □

The prize is therefore a concrete σ-complete OML with a σ-additive state outside the dispersion-free hull.

Why σ-essential. The finite version of this is already inhabited. Wright's pentagon [40] is a finite, concrete OML — a Greechie loop of five three-atom blocks, hence a lattice — with an order-determining family of dispersion-free states, yet it carries a non-classical $\{0, \frac{1}{2}\}$-valued state lying outside their closed convex hull. (The five intertwining atoms form a 5-cycle in which consecutive atoms share a block, so any dispersion-free state, taking at most one of each adjacent pair to 1, sets at most 2 of the 5 to 1; their hull has vertex-sum ≤ 2, whereas the $\{0, \frac{1}{2}\}$ state — a legitimate state, each block summing to 1 — has vertex-sum $\frac{5}{2}$. This is the KCBS inequality [24].) So "concrete OML with a contextual state" is a 1978 fact, and the general spanning statement is false. What Wright does *not* supply is contextuality that requires the countable structure: the pentagon's is witnessed by a finite sub-OML (itself). The prize must therefore be *σ-essential* — contextual in the σ-complete whole but in no finite sub-OML. This is not a device to evade Wright: "every finite piece has a global section, the countable whole has none" is a *compactness failure* of exactly the type by which countable additivity is not first-order axiomatizable over finitely additive coherence (see Remark 5.4).

Status. The σ-essential cell has *no known inhabitant* and *no impossibility proof*. The product $\prod_n \mathrm{MO}_2$ fails it: its contextual states are only finitely additive — diffuse states on its central $P(\mathbb{N})$, which σ-additivity forces to concentrate on points, killing the contextuality (§5.2); Wright is finite; no other construction is known. The question is well-posed but uninhabited.

Remark 5.4 (The compactness-failure analogy). The σ-essential requirement places the open problem alongside a known Boolean phenomenon. Over a Boolean algebra, countable additivity of a finitely additive charge is not first-order axiomatizable: an ultraproduct of σ-additive charges can be finitely additive but not σ-additive, so every finite (indeed first-order) test is passed while the global property fails [27]. A σ-essential contextual state would be the non-distributive analogue — finite sub-OMLs all classically interpretable, the obstruction carried only by the countable whole.

5.2. **What the combinatorial route settles, and why it is not enough.** It is tempting to look for the object among concrete σ-orthocomplete OMLs, on the grounds that a state's σ-additivity invokes $\bigvee_n a_n$ only for orthogonal families (Definition 2.5). This route is **settled, and it does not produce**

the object. The property that would make σ-additivity genuinely exceed finite additivity on a concrete non-Boolean lattice is the interleaving condition

$$(\bigstar)\quad \exists\, p,\ \exists\,\{a_n\} \text{ infinite orthogonal},\quad p\wedge a_n=0\ (\forall n),\quad p\not\leq a_n\ (\forall n), \tag{2}$$

and $(\bigstar)$ is satisfied *trivially* by the plain product $\prod_n \mathrm{MO}_2$: take $a_n = a$ in coordinate n (zero elsewhere), and $p = \bigvee_n b_n$ where $b_n = b$ in coordinate n (zero elsewhere) — a genuine countable *orthogonal* join, so $p = (b, b, b, \dots)$ exists by σ-orthocompleteness. Then $p \wedge a_n = 0$ (coordinate n: $a \wedge b = 0$ in MO_2; elsewhere $a_n = 0$) and $p \not\leq a_n$ (coordinate n: $b \not\leq a^\perp$). Since $\prod_n \mathrm{MO}_2$ is concrete, σ-complete (a product of finite, hence complete, lattices), non-Boolean, and infinite, it satisfies $(\bigstar)$ and *every* combinatorial hypothesis one might impose — yet it is exactly the *segregated* object the open problem must exclude. So the combinatorial axioms (σ-completeness, concreteness, $(\bigstar)$) are *not* the discriminator; the missing ingredient is non-segregation, which is a point-free notion, not a closure property.

Remark 5.5 (The witness has no points, and the Boolean boundary)*.* The universal closure above (a faithful set representation forces distributivity) already rules out the point-based route for every non-Boolean OML. The witness $\prod_n \mathrm{MO}_2$ shows the failure concretely and at the level of points: MO_2 admits *no* two-valued *homomorphism* — each of its four separating two-valued *states* ($a \mapsto 0$, $b \mapsto 0$, etc.) fails $\omega(a \vee b) \leq \omega(a) + \omega(b)$ since $a \vee b = \mathbf{1}$, so none preserves $\vee$ — and $\prod_n \mathrm{MO}_2$ inherits this, since its diagonal copy $\{\mathbf{0}, \mathbf{1}, (a, a, \dots), (a^\perp, \dots), (b, b, \dots), (b^\perp, \dots)\}$ is a sub-MO_2 on which any homomorphism would restrict to one of MO_2's (none exists). (This is special to MO_2 and its products — a generic non-Boolean OML such as $\mathbf{2} \times \mathrm{MO}_2$ *does* carry homomorphisms, e.g. the projection; the universal obstruction is the distributivity argument, not homomorphism-scarcity.) So the witness is concrete (Definition 2.7: order-determining two-valued *states*) and σ-complete yet has no points in the homomorphism sense at all — the point-free target is forced, not engineered. The same arithmetic records the sharp Boolean boundary: an OML is Boolean iff it admits a *unital* set of *subadditive* states — one with, for each nonzero a, a subadditive state giving a value 1 (Pták–Pulmannová [33], for OMLs). A concrete non-Boolean OML therefore cannot have its separating two-valued states form such a unital set: some fail subadditivity (here all four of MO_2's fail $\omega(a \vee b) \leq \omega(a) + \omega(b)$). The forcing property to Boolean is thus subadditivity — a finitary condition — not σ-additivity.

A worked instance fixes the picture. Navara's construction [30] produces an infinite, σ-orthocomplete, rich OML whose inner block is a finite *stateless* Greechie logic — which is exactly what makes Navara's $\mathcal{L}$ non-concrete. Replacing the stateless block by MO_2 (write $\mathcal{L}_2$) removes the only source of non-concreteness: the assembly is a product of a sublogic of a Kalmbach horizontal sum (blocks glued only at $\{0, 1\}$, *not* an atom-sharing Greechie loop), and pure horizontal sums of concrete logics are concrete, as are sublogics and products. So $\mathcal{L}_2$ is concrete, σ-orthocomplete, non-Boolean, and carries ($\bigstar$) — yet it is not a new object: it is $\prod_n \mathrm{MO}_2$ with extra scaffolding, segregated in the same way, and it likewise admits no σ-tribe representation. It is the cautionary example showing that exhibiting ($\bigstar$) and concreteness together is easy and does *not* reach the open problem.

The art constrains the open (σ-complete) problem from both sides without closing it: *no* impossibility theorem covers the concrete class (Remark 4.1), while the two *theorem-closed* routes to an engine — RDP and MacNeille (§4.3(i),(ii)) — are ruled out, leaving only the third, σ-Stone duality, which is itself open. What is required is not a new idea in the abstract but a representation in that gap: non-distributive enough to evade routes (i),(ii), yet σ-faithful where the known dualities are merely finitary.

Remark 5.6 (Why $L(H)$+Gleason does *not* already settle this — the (A)/(B) point-space equivocation). "Probability from relations, point-free" may *look* delivered by $L(H)$+Gleason. It is not: the appearance rests on conflating two point-spaces, and *realisation* is defined as concentration on the second. The two are (A) the Hilbert rays of H, which Gleason *requires* — the density ρ in $s = \mathrm{tr}(\rho\,\cdot)$ is built from them — and (B) the McDonald–Bimbó dual filters $P(A)$, which Gleason *removes*, but only by being a representation theorem. Gleason is thus the *most* point-ful result available (every lattice state corresponds to the point datum ρ). What $L(H)$ exhibits is a *separation*, not point-freeness: σ-additive measures exist on the lattice (Gleason, for normal states), yet on the dual $S_0(L(H))$ no state extends to a charge at all (Proposition 3.2), so a fortiori none concentrates on $P(A)$. It cannot serve as the existence proof required, so the point-free σ-additive structure of §5 is the only candidate engine.

5.3. **Two axes, two dictionaries — and the cell with no dictionary.** The recurring confusion of this subject (Remarks 5.5, 5.6) is best held as a 2×2 on two *independent* axes, which must not be collapsed into one.

- **Axis 1 — distributivity = commutativity.** One condition in two dialects: a family of projections commutes *iff* the lattice they generate is distributive. "Commutativity" is the operator word, "distributivity" the lattice word, for the *same* dividing line — not different sides.
- **Axis 2 — concreteness.** A non-Boolean lattice can be *point-rich in rays* (atoms = 1-dim subspaces) yet *point-free in valuations* (no two-valued homomorphisms, Kochen–Specker), or the reverse: the two senses of "point" come apart (the (A)/(B) equivocation, Rmk 5.6).

	distributive (commuting)	**non-distributive (non-commuting)**
concrete (valuations span)	classical: Boolean σ-algebra, Loomis–Sikorski free	the descent witness lives here: a concrete σ-complete non-Boolean OML ($\prod_n \mathrm{MO}_2$ lies here, but segregated)
non-concrete (KS: few valuations)	atomless measure algebra (valuation-poor)	$L(H)$: rays abundant, valuations KS-barred — the operator / e.s.a. world, the excluded pole

There is *not* one "lattice $\leftrightarrow$ space" dictionary but two, one per row. *Birkhoff–von Neumann* (bottom-right): elements $\leftrightarrow$ closed subspaces, atoms $\leftrightarrow$ **rays**, $a \vee b \leftrightarrow$ closed span (the overshoot — a 2-plane holds rays in neither joinand: the geometric face of the meet/join gap, Def. 2.2), compatibility $\leftrightarrow$ commutativity; rich in ray-points, KS-barred from valuation-points, hence *non-concrete*. *Stone / concrete-logic* (Gudder; top row): elements $\leftrightarrow$ sets of outcomes, atoms $\leftrightarrow$ points of Ω (genuine valuation points), orthogonal $\vee \leftrightarrow$ disjoint union (only *incompatible* joins overshoot); rich in valuation-points, hence *concrete*. The operator question and the descent question share **Axis 1** (same column) but sit in **opposite rows**: operators *force* the bottom-right cell (spectral projections $\Rightarrow L(H) \Rightarrow$ KS $\Rightarrow$ non-concrete), whereas the descent witness must be **top-right**. So the descent problem is the one cell *no operator algebra can occupy* — which is why the witness must be a pasting / countable colimit of concrete blocks, not a Hilbert-space construction, and why $L(H)$+Gleason (Rmk 5.6) cannot settle it. In this language the open problem (route (iii), §4.3) is exactly: *is there a logic $\leftrightarrow$ geometry dictionary for the top-right cell at all?* — Birkhoff–von Neumann covers bottom-right, Stone covers the concrete distributive corner, and a σ-Loomis–Sikorski for concrete OMLs *is* the missing top-right dictionary. (Orientation, not a new result; the danger it guards is collapsing Axis 2 into Axis 1 — treating "non-distributive" as if it meant "$L(H)$" — which re-merges the two poles concreteness separates.)

6. Directions

With the extension axis closed on $L(H)$ (Proposition 3.2), the combinatorial route shown to deliver only segregated objects (§5.2), and the prize reduced to a σ-essential contextual state (§5.1), the problem has a clean dichotomy: two exits, each a genuine terminus.

Exhibit a σ-essential witness: Construct an infinite concrete OML carrying a σ-additive contextual state that no finite sub-OML witnesses. The natural attack is a compactness-failure construction (cf. Remark 5.4): a concrete σ-complete OML whose finite sublogics are all classically interpretable (dispersion-free states span them) but whose countable joins force a Kochen–Specker-type obstruction only in the limit. The faithful set representation is ruled out at every stage (Theorem 5.2), so the obstruction must be carried by the σ-structure itself, not by any finite block. A necessary condition follows: the witness must place its infinitary structure *off-center* (irreducible, with non-central infinite blocks). A product of finite blocks such as $\prod_n \mathrm{MO}_2$ is excluded a priori, since it forces all infinitary content into the Boolean center, where σ-additivity concentrates the contextual states onto points and removes them (§5.2); a pasting or countable colimit of Wright-type blocks is the natural place to look.

Prove no such witness exists:
A spanning theorem: every σ-additive state on a σ-complete concrete OML lies in the closed convex hull of its dispersion-free states *except* where a finite sub-OML is already contextual (Wright). This would be an impossibility result — a σ-essential contextual state cannot exist, so relational σ-additive probability is operationally invisible (present finitely or not at all). The Boolean boundary — subadditivity of a unital set of states (Pták–Pulmannová, Remark 5.5) — is the natural lever. Note that such an argument cannot reuse the $\prod_n \mathrm{MO}_2$ mechanism (concentration on the Boolean center): that is vacuous off-center, so a genuinely different finite-witnessing argument is required.

A candidate direction now closed: the singular case for $L(H)$. One might have hoped a singular state would extend where no normal state does, providing the first explicit "extends but may fail to concentrate" example; Proposition 3.2 rules this out, catching every state, singular included. The reduction is recorded because the singular case was expected to require infinitary methods and does not (Remark 6.1).

Remark 6.1 (Finite versus countable: finite suffices). Is the obstruction to extending a singular state realised by a *finite* or only by a *countable* pairwise-meet-zero family? The two arguments divide on exactly this point. The line argument (Remark 3.3) is finitary but *vacuous* on a singular state ($s(e) = 0$ for every finite-rank e); the MO_2 argument with infinite-dimensional atoms (Proposition 3.2) uses a finite family ($n = 4$) and gives $\sum_i s(a_i) = 2$ state-independently, catching the singular case as well. So the singular obstruction is *not* entangled with the descent-axis passage, as the vacuity of the line argument might suggest: it is realised by a finite family, hence a self-contained finitary problem, separable from the general construction and not governed by the finitary-to-σ passage.

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